

V Semester			
S. No	Course Code	Course Title	Credit Hours
1.	AECON-311	Agricultural Marketing Trade	3(2+1)
2.	AGRON-311	Introduction to Agro-meteorology	2(1+1)
3.	AGRON-312	Weed Management	2(1+1)
4.	ENTO-311	Pests Management in Crops and Stored Grains	3(2+1)
5.	AENGG-311	Renewable Energy in Agriculture and Allied Sector	2(1+1)
6.	GPB-311	Crop Improvement (<i>Kharif</i> crops)-I	2(1+1)
7.	HORT-311	Ornamental Crops, MAP and Landscaping	2(1+1)
8.	PPATH-311	Diseases of Field & Horticultural Crops and their Management	3(2+1)
9.	PPHYS-311	Fundamentals of Crop Physiology	3(2+1)
10.	ET-311	Educational Tour**	2(0+2)
Total Credit			22(13+9)+2**
**NC: Non-gradual courses			

AECON-311	Agricultural Marketing Trade	3(2+1)
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Objective

1. To understand the fundamentals of agricultural marketing and trade
2. To analyze the factors influencing supply and demand in agricultural markets
3. To explore different marketing channels and strategies in agriculture
4. To examine the role of government policies and regulations in agricultural markets

Theory

Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing, market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets; Demand, supply and producer's surplus of agri commodities: nature and determinants of demand and supply of farm products, producer's surplus– meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agri-commodities; Pricing and promotion strategies: pricing considerations and approaches–cost based and competition based pricing; market promotion – advertising, personal selling, sales promotion and publicity – meaning, merits and demerits; Marketing process and functions: Marketing process concentration, dispersion and equalization; exchange functions – buying and selling; physical functions–storage, transport and processing; facilitating functions– packaging, branding, grading, quality control and labeling (Agmark); Market functionaries and marketing channels: Types and importance of agencies involved in agricultural

marketing; meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products; Integration, efficiency, costs and price spread: Meaning, definition and types of market integration; marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing; reasons for higher marketing costs of farm commodities; ways of reducing marketing costs; Role of Govt. in agricultural marketing: Public sector institutions- CWC, SWC, FCI, CACP and DMI – their objectives and functions; cooperative marketing in India; Risk in marketing: Types of risk in marketing; speculation and hedging; an overview of futures trading; Agricultural prices and policy: Meaning and functions of price; administered prices; need for innovations in agricultural price policy; Trade: Concept of International Trade and its need, theories of absolute and comparative advantage. Present status and prospects of international trade in agri-commodities; WTO; Agreement on Agriculture (AoA) and its implications on Indian agriculture; IPR; Role of government in agricultural marketing; Role of APMC and its relevance in the present day context.

Practical

Plotting and study of demand and supply curves and calculation of elasticities; Study of relationship between market arrivals and prices of some selected commodities; Computation of marketable and marketed surplus of important commodities; Study of price behaviour over time for some selected commodities; Construction of index numbers; Visit to a local market to study Various marketing functions performed by different agencies, identification of marketing channels for selected commodity, collection of data regarding marketing costs, margins and price spread and presentation of report in the class; Visit to market institutions–NAFED, SWC, CWC, cooperative marketing society, etc. to study their organization and functioning. Application of principles of comparative advantage of international trade.

Lecture Schedule: Theory

S. No.	Name of Topic	No. of Lectures
1.	Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing	1
2.	Market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets	2
3.	Demand, supply and producer's surplus of agri commodities: nature and determinants of demand and supply of farm products	2
4.	Producer's surplus– meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agri-commodities	1
5.	Pricing and promotion strategies: pricing considerations and approaches–cost	2

	based and competition based pricing; market promotion – advertising, personal selling, sales promotion and publicity – meaning, merits and demerits	
6.	Marketing process and functions: Marketing process concentration, dispersion and equalization; exchange functions – buying and selling;	2
7.	Physical functions–storage, transport and processing; facilitating functions–packaging, branding, grading, quality control and labeling (Agmark)	2
8.	Market functionaries and marketing channels: Types and importance of agencies involved in agricultural marketing; meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products	2
9.	Integration, efficiency, costs and price spread: Meaning, definition and types of market integration	2
10.	Marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing; reasons for higher marketing costs of farm commodities; ways of reducing marketing costs	2
11.	Role of Govt. in agricultural marketing: Public sector institutions- CWC, SWC, FCI, CACP and DMI – their objectives and functions; cooperative marketing in India	3
12.	Risk in marketing: Types of risk in marketing	1
13.	Speculation and hedging; an overview of futures trading	2
14.	Agricultural prices and policy: Meaning and functions of price; administered prices; need for innovations in agricultural price policy	2
15.	Trade: Concept of International Trade and its need, theories of absolute and comparative advantage	2
16.	Present status and prospects of international trade in agri-commodities; WTO; Agreement on Agriculture (AoA) and its implications on Indian agriculture	2
17.	IPR; Role of government in agricultural marketing; Role of APMC and its relevance in the present day context	2

Lecture Schedule: Practical

S. No.	Name of Topic	No. of Lectures
1.	Plotting and study of demand and supply curves	1
2.	Calculation of elasticities	1
3.	Study of relationship between market arrivals and prices of some selected commodities	1

4.	Computation of marketable and marketed surplus of important commodities	2
5.	Study of price behaviour over time for some selected commodities	2
6.	Construction of index numbers	1
7.	Visit to a local market to study Various marketing functions performed by different agencies, identification of marketing channels for selected commodity, collection of data regarding marketing costs, margins and price spread and presentation of report in the class	3
8.	Visit to market institutions–NAFED, SWC, CWC, cooperative marketing society, etc. to study their organization and functioning	3
9.	Application of principles of comparative advantage of international trade	2

Suggested readings

1. Acharya, S. S. and Agarwal, N.L. 2006. Agricultural Marketing in India, Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
2. Chinna, S. S. 2005. Agricultural Economics and Indian Agriculture. Kalyani Pub, N Delhi.
3. Dominic Salvatore, Micro Economic Theory
4. Kohls Richard, L. and Uhl Josheph, N. 2002. Marketing of Agricultural Products, Prentice-Hall of India Private Ltd., New Delhi.
5. Kotler and Arm strong, 2005. Principles of Marketing, Pearson Prentice-Hall.
6. Lekhi, R. K. and Joginder Singh. 2006. Agricultural Economics. Kalyani Publishers, Delhi.
7. Memoria, C. B., Joshi, R. L. and Mulla, N. I. 2003. Principles and Practice of Marketing in India, Kitab Mahal, New Delhi.
8. Pandey Mukesh and Tewari, Deepali. 2004. Rural and Agricultural Marketing, International Book Distributing Co. Ltd, New Delhi.
9. Sharma, R. 2005. Export Management, Laxmi Narain Agarwal, Agra.

AGRON-311	Introduction to Agro-meteorology	2(1+1)
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Objectives

1. To introduce the students to the concept of weather and climate and underlying physical processes occurring in relation to plant and atmosphere
2. To impart the theoretical and practical knowledge of instruments/equipment used for measurement of different weather variables in an agrometeorological observatory
3. To study the meteorological aspects of climate change in agriculture and allied activities

Theory

Meaning and scope of agricultural meteorology; Earth atmosphere- its composition, extent and structure; Atmospheric weather variables; Atmospheric pressure, its variation with height; Wind, types of wind, daily and seasonal variation of wind speed, cyclone, anticyclone, land breeze and sea breeze; Nature and properties of solar radiation, solar constant, depletion of solar radiation, short wave, long wave and

thermal radiation, net radiation, albedo; Atmospheric temperature, temperature inversion, lapse rate, daily and seasonal variations of temperature, vertical profile of temperature, Application of Thermal time concept and Crop/Pest weather calendar; Energy balance of earth; Atmospheric humidity, concept of saturation, vapor pressure, process of condensation, formation of dew, fog, mist, frost, cloud; Precipitation, process of precipitation, types of precipitation such as rain, snow, sleet, and hail, cloud formation and classification; Artificial rainmaking. Monsoon- mechanism and importance in Indian agriculture, Weather hazards - drought, floods, frost, tropical cyclones and extreme weather conditions such as heat-wave and cold-wave. Agriculture and weather relations; Modifications of crop microclimate, climatic normals for crop and livestock production. Weather forecasting- types of weather forecast and their uses. Climate change, climatic variability, global warming, causes of climate change and its impact on regional and national Agriculture.

Practical

Visit of Agro meteorological Observatory, site selection of observatory, exposure of instruments and weather data recording. Measurement of total, shortwave and longwave radiation, and its estimation using Planck's intensity law. Measurement of albedo and sunshine duration, computation of Radiation Intensity using BSS. Measurement of maximum and minimum air temperatures, its tabulation, trend and variation analysis. Measurement of soil temperature and computation of soil heat flux. Determination of vapor pressure and relative humidity. Determination of dew point temperature. Measurement of atmospheric pressure and analysis of atmospheric conditions. Measurement of wind speed and wind direction, preparation of wind rose. Measurement, tabulation and analysis of rain. Measurement of open pan evaporation and evapotranspiration. Computation of PET and AET

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Meaning and scope of agricultural meteorology	1
2.	Earth's atmosphere- its composition, extent and structure	1
3.	Atmospheric weather variables; Atmospheric pressure, its variation with height	1
4.	Wind, types of wind, daily and seasonal variation of wind speed	1
5.	Cyclone, anticyclone, land breeze and sea breeze	1
6.	Nature and properties of solar radiation, solar constant, depletion of solar radiation	1
7.	Short wave, long wave and thermal radiation, net radiation, albedo	1

8.	Atmospheric temperature, temperature inversion, lapse rate, daily and seasonal variations of temperature, vertical profile of temperature, Application of Thermal time concept and Crop/Pest weather calendar	1
9.	Energy balance of earth; Atmospheric humidity, concept of saturation, vapor pressure,	1
10.	Process of condensation, formation of dew, fog, mist, frost, cloud	1
11.	Precipitation- process , types such as rain, snow, sleet, and hail	1
12.	Cloud formation and classification; Artificial rainmaking, Monsoon-mechanism and importance in Indian agriculture	1
13.	Weather hazards - drought, floods, frost, tropical cyclones and extreme weather conditions such as heat-wave and cold-wave	1
14.	Agriculture and weather relations; Modifications of crop microclimate, climatic normals for crop and livestock production	1
15.	Weather forecasting- types of weather forecast and their uses	1
16.	Climate change, climatic variability, global warming, causes of climate change and its impact on regional and national Agriculture	1

Lecture schedule: Practical

S.N.	Topic	No. of lectures
1.	Visit of Agrometeorological Observatory, site selection of observatory, exposure of instruments and weather data recording	2
2.	Measurement of total, shortwave and long wave radiation, and its estimation using Planck's intensity law	2
3.	Measurement of albedo and sunshine duration, computation of Radiation Intensity using BSS	2
4.	Measurement of maximum and minimum air temperatures, its tabulation, trend and variation analysis	1
5.	Measurement of soil temperature and computation of soil heat flux	1
6.	Determination of vapor pressure and relative humidity	1
7.	Determination of dew point temperature	1

8.	Measurement of atmospheric pressure and analysis of atmospheric conditions	1
9.	Measurement of wind speed and wind direction, preparation of wind rose.	1
10.	Measurement, tabulation and analysis of rain	1
11.	Measurement of open pan evaporation and evapotranspiration	1
12.	Computation of PET and AET	1
13.	Use of synoptic charts, weather reports, weather forecasting-types and methods, crop weather calendar	1

Suggested Readings

1. Agricultural Meteorology by G.S.L.H.V. Prasado Rao
2. Fundamentals of Agrometeorology and Climate Change by G. S. Mahi and P. K. Kingra
3. Introduction to Agrometeorology and Climate Change by Alok Kumar Patra
4. Introduction to Agrometeorology by H. S. Mavi
5. Text Book of Agricultural Meteorology by M. C. Varshneya and P.B. Pillai

AGRON-312	Weed Management	2(1+1)
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Objectives

1. To teach students about principles of weed science
2. To impart practical knowledge of weed management in field and horticultural crops

Theory

Introduction to weeds, characteristics of weeds, their harmful and beneficial effects on ecosystem. Classification, reproduction and dissemination of weeds, crop-weed competition, factors of competition, factors affecting growth and development. Studies on weed seed bank, weed shifts. Concepts of weed management: physical, cultural, chemical and biological; principles and methods, integrated weed management. Implements for weed control, robotic weed control, weed management in organic/ natural farming. Herbicide classification and properties of important herbicides, concept of adjuvants, surfactants, herbicide formulation and their use, Nano herbicides, precision weed management; Mode of action of herbicides and selectivity phenomenon. Concept of herbicide mixture and utility in agriculture, Herbicide compatibility with agro-chemicals and their application, Herbicide resistance and its management. Weed management in different field and horticultural crops; aquatic weed management, weed management in cropping systems.

Practical

Techniques of weed preservation, weed identification and losses caused by weeds. Biology of important weeds. Study weeds in different situations, Study of herbicide formulations and mixture of herbicide. Study methods of herbicide application, Herbicide application equipment their parts, use, maintenance

and calibration. Weed control implements, Calculation of herbicide doses and requirement, weed control efficiency and weed index, Phytotoxicity of herbicides, Weed management in fallow lands, Management of problem and parasitic weeds.

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Introduction to weeds, Characteristics of weeds	1
2.	Harmful and beneficial effects of weeds on ecosystem	1
3.	Classification of weeds	1
4.	Reproduction and dissemination of weeds	1
5.	Crop-weed competition, factors of competition, factors affecting growth and development, Studies on weed seed bank, weed shifts	1
6.	Concepts of weed management: physical, cultural, chemical and biological; principles and methods, integrated weed management	2
7.	Implements for weed control, robotic weed control, Weed management in organic/ natural farming	1
8.	Herbicide classification and properties of important herbicides	1
9.	Concept of adjuvants, surfactants, herbicide formulation and their use	1
10.	Nano herbicides, precision weed management	1
11.	Mode of action of herbicides and selectivity phenomenon	1
12.	Concept of herbicide mixture and utility in agriculture, Herbicide compatibility with agro-chemicals and their application	1
13.	Herbicide resistance and its management	1
14.	Weed management in different field and horticultural crops;	1
15.	Aquatic weed management, Weed management in cropping systems	1

Lecture schedule: Practical

S.N.	Topic	No. of lectures
1.	Identification of weeds and techniques of weed preservation	2
2.	Collection of common <i>kharif/rabi</i> weeds and their preservation	1
3.	Collection of common perennial weeds and their preservation	1
4.	Biology of important weeds	1
5.	Study weeds in different situations	1
6.	Study of herbicide formulations and mixture of herbicide.	2

7.	Study methods of herbicide application	1
8.	Herbicide application equipment their parts, use, maintenance and calibration	1
9.	Weed control implements	1
10.	Calculation of herbicide doses and requirement	1
11.	Calculations of weed control efficiency and weed index	1
12.	Phytotoxicity of herbicides	1
13.	Weed management in fallow lands	1
14.	Management of problem and parasitic weeds	1

Suggested Readings

1. Crafts, A.S. and Robbins, W.W. 1973. Weed Control. Tata McGraw-Hill Publishing Co. Ltd., New Delhi.
2. Gupta, O.P. 1984. Scientific Weed Management. Today and Tomorrow Printers and Publishers, New Delhi.
3. Gupta, O.P. 2015. Modern Weed Management. Agro Bios (India), Jodhpur.
4. Naidu, V.S.G.R. Handbook of Weed Identification. Directorate of Weed Research, Jabalpur.
5. Rajagopal, A., Aravindan, R. and Shanmugavelu, K.G. 2015. Weed management of Horticultural Crops. Agrobios (India), Jodhpur.
6. Ramamoorthy, K. and Subbian, P. Predominant Weed flora in hill –ecosystems. Agrobios (India), Jodhpur.
7. Rao, V.S. 2000. Principles of Weed Science. Oxford & IBH Publishing Co., New Delhi
8. Subramanian, S., Mohammed Ali, A. and Jayakumar, R. 1991. All About Weed Control. Kalyani Publishers, Ludhiana.
9. Tadulingam, C. and Venkatnarayana, D. 1955. A Handbook of Some South Indian Weeds. Government Press, Madras.
10. Thakur, C. 1977. Weed Science. Metropolitan Book Co. Pvt. Ltd., New Delhi

ENTO-311	Pests Management in Crops and Stored Grains	3(2+1)
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Objectives

Diagnosis and management of major insect and non-insect pests of crops in field and storage

Theory

Concept of IPM, Practices, scope and limitations of IPM. Classification of insecticides, toxicity of insecticides and formulations of insecticides, Biorational pesticides including insect repellents, anti feedants, Use of drones and AI in pest management. General description on nature and type of damage by different arthropod pests; Scientific name, order, family, host range, distribution, biology and bionomics; Nature of damage and management of major insect pests of various field crops, vegetable crops, fruit crops, plantation crops, ornamental crops, spices and condiments. Invasive insect pests and their management. Insect pests of stored grains, rodents and their management. Storage structures and methods of grain storage and fundamental principles of stored grains management.

Practical

Field visit, identification of major insect pests and their damage symptoms. Collection and preservation of major insect pests; collection of damage samples, their identification and herbarium preparation. Methods of monitoring of pest incidence *in situ*. Management strategies of insect pests of different crops. Study on structural entomology and household pests. Storage structures and methods of grain storage. Structure and function of sprayer and duster. Spraying techniques for selected field and horticultural crops. Vertebrate pest management, Mass multiplication of NPV.

Theory

S.No.	Topic	No. of lecture
1.	Concept of IPM, practices, scope and limitations of IPM.	1
2.	Classification of insecticides, toxicity of insecticides and formulations of insecticides.	2
3.	Biorational pesticides including insect repellents, antifeedants, Use of drones And AI in pest management.	2
4.	Polyphagous insect pests: Locust, grasshopper, white grub, termite and red Hairy caterpillar	2
5.	Rice: Brown plant hopper and yellow stem borer.	1
6.	Sorghum: Shootfly; Maize: Stem borer; Sugarcane: Pyrilla, whitefly, shoot borer	2
7.	Pulses: Gram pod borer, cutworm	1
8.	Oilseeds: Mustard aphid, sawfly, painted bug, soybean girdle beetle, sesame Leaf and capsule borer.	2
9.	Cotton: Jassid, whitefly, spotted and pink bollworm.	2
10.	Pests of vegetables crops: Brinjal- brinjal shoot and fruit borer; Tomato- Fruit borer (Covered in gram); Okra- Shoot and fruit borer (Covered in cotton).	2
11.	Chilli: Thrips; Onion and garlic: Thrips	1
12.	Cruciferous vegetables: Cabbage caterpillar, diamond back moth, tobacco caterpillar.	1
13.	Cucurbitaceous vegetables: Melon fruitfly, red pumpkin beetle, red vegetable mite.	1
14.	Pests of fruit crops: Mango: Mango hopper, mealy bug, fruitfly; Guava: Fruitfly.	1
15.	Citrus: Citrus psylla, citrus caterpillar, bark eating caterpillar.	1

16.	Pomegranate:Anar butterfly; Ber:Fruitfly.	1
17.	Coconut: Rhinoceros beetle;Red palm weevil, Apple:woolly aphid.	1
18.	Ornamental Crops:Rose aphid	1
19.	Spices and condiments:Aphid	1
20.	Invasive insect pests and their management.	2
21.	Insect pests of stored grains, rodents and their management.	2
22.	Storage structures and methods of grain storage.	2

Practical

S.No.	Topic	No. of lecture
1.	Identification of major insect pests and their damage symptoms.	2
2.	Collection and preservation of major insect pests	2
3.	Collection of damage samples, their identification and herbarium preparation	2
4.	Methods of monitoring of pest incidence <i>in situ</i>	1
5.	Management strategies of insect pests of different crops.	2
6.	Storage structures and methods of grain storage.	2
7.	Structure and function of sprayer and duster.	1
8.	Spraying techniques for selected field and horticultural crops.	2
9.	Vertebrate pest management,	1
10.	Mass multiplication of NPV	1

Suggested readings

1. A Textbook of Insect Pest and Disease Management, 2021. Somnath Sen, and Mohd. Sameer, S. Kataria & Sons publish.
2. Agricultural Pests of India and Southeast Asia, A.S. Athwal, Kalyani Publishers.
3. A Textbook of Applied Entomology, K.P. Srivastava and G. S. Dhaliwal, Kalyani Publish.
4. Essentials of Pest Management: Key Information on Pest Identification and its Management, 2022. Prakash Rambhat Thalya and Ravi Chandra
5. Integrated pest Management Concept and Approaches-G.S.Dhaliwal and Ramesh Arora
6. Pest Management: Methods, Applications and Challenges, Tarique Hassan Askary, Agriculture, Agriculture Issues and policies, Books, Nova, Pest Control, Science and Technology, 2022

AENGG-311	Renewable Energy in Agriculture and Allied Sector	2(1+1)
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Objectives

1. To gain the knowledge on different types of materials used in Renewable Energy.
2. To understand the importance of Renewable Energy technology and its applications.
3. To train the students on the applications of solar thermal technology.

Theory

Classification of energy sources, contribution of these of sources in agricultural sector; Familiarization with biomass utilization for bio-fuel production and their application; Familiarization with types of biogas plants and gasifiers, biogas, bio-alcohol, biodiesel and bio-oil production and their utilization as bioenergy resource; Introduction of solar energy, collection and their application; Familiarization with solar energy gadgets: solar cooker, solar water heater, application of solar energy: solar drying, solar pond, solar distillation, solar photovoltaic system and their application; Introduction of wind energy and their application. Availability of bio mass and their application in different places.

Practical

Familiarization with renewable energy gadgets. To study biogas plants, gasifier, production process of biodiesel, briquetting machine, production process of bio-fuels. Familiarization with different solar energy gadgets. To study solar photovoltaic system: solar light, solar pumping, solar fencing, solar cooker and solar drying system. To study solar distillation, solar pond and solar wind hybrid system. Field visit to Solar –Wind farm.

Lecture schedule: Theory

S. No.	Topic	No. of lectures
1.	Classification of energy sources, contribution of these sources in agricultural sector.	1
2.	Familiarization with biomass utilization for bio fuel production and their application.	2
3.	Familiarization with types of biogas plants and gasifiers,	2
4.	Biogas, bio-alcohol, biodiesel and bio-oil production and their utilization as bio-energy resource.	2
5.	Introduction of solar energy, collection and their application;	2
6	Familiarization with solar energy gadgets: solar cooker, solar water heater.	1
7	Application of solar energy: solar drying, solar pond, solar distillation.	2
8	Solar photovoltaic system and their application	2
9	Introduction of wind energy and their application.	1
10	Availability of bio mass and their application in different places.	1

Lecture schedule: Practical

S.NO.	Topic	No. of lectures
1	Study of fixed dom and floating drum type biogas plants.	2
2	Study of cross draft, updraft and down draft gasifiers.	2
3.	Study of Production process of bio-diesel.	1
4	To study briquetting machine.	1
5	Study of production process of bio-fuels.	1
6	Study of solar water heating and solar distillation system.	1
7	Study of solar dryer, solar cooker and animal concentrate cooker.	1
8	Study of solar photovoltaic systems for lighting and solar fencing.	2
9	Study of solar photovoltaic water pumping system and visit to nearby solar	1
10	Study of solar photovoltaic sprayer and duster.	1
11	Study of solar pond.	1
12	Study of wind mill and solar wind hybrid system.	1
13	Field visit to Solar –Wind farm.	1

Suggested readings

1. C.S. Solanki. 2011. Solar Photovoltaic – Fundamentals, Technologies and Applications. PHI Learning Pvt. Ltd.
2. S. Sukhatme and J. Nayak. 2008. Solar Energy: Principles of Thermal Collection and Storage. Third Edition (Tata McGraw-Hill).
3. G.D. Rai. Non-Conventional Energy Sources, Kh Publishers, New Delhi.
4. N.S. Rathore. A.K. Kurchania, N.L. Panwar. (2007). Non Conventional Energy Sources, Himanshu Publications.
5. N.S. Rathore. A. K. Kurchania, N.L. Panwar. (2007). Renewable Energy, Theory and Practice, Himanshu Publications.
6. K.C. Khandelwal. & S.S. Mandi. (1990). Biogas Technology.

GPB-311	Crop Improvement (<i>Kharif</i> crops)-I	2(1+1)
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Objectives

1. To provide knowledge about Self-pollinated and cross pollinated *Kharif* crops
2. To learn about origin and distribution of *Kharif* crops
3. To design breeding objectives of major *kharif* crops
4. To impart information on different crop varieties for *Kharif* season

Theory

Centres of origin, distribution of species, wild relatives in different cereals; pulses; oilseeds; fibres; fodders and cash crops; vegetable and other horticultural crops of *kharif* season; Plant genetic resources, its utilization and conservation, study of genetics of qualitative and quantitative characters; Important concepts of breeding self-pollinated, cross-pollinated and

vegetatively propagated crops. Major breeding objectives and procedures including conventional and modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional); Hybrid seed production technology in maize, rice, sorghum, pearl millet and pigeonpea etc. Ideotype concept, climate resilient crop varieties for future.

Practical

Botany of crops, Floral biology, emasculation and hybridization techniques in different crop species, viz. rice, jute, maize, sorghum, pearl millet, ragi, pigeonpea, urdbean, mungbean, soybean, groundnut, sesame, castor, cotton, cowpea, tobacco, brinjal, okra and cucurbitaceous crops. Maintenance breeding of different *kharif* crops. Handling of germplasm and segregating populations by different methods like pedigree, bulk and single seed decent methods; Study of field techniques for seed production and hybrid seed production in *kharif* crops; Estimation of heterosis, inbreeding depression and heritability; Layout of field experiments; Study of quality characters, donor parents for different characters; Visit to seed production plots; Visit to AICRP breeding plots of different crops.

Lecture Schedule: Theory

S.N.	Topic	No. of lectures
1	Crop improvement aspects in rice as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc & hybrid seed production	1
2	Crop improvement aspects in maize as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc & hybrid seed production	1
3	Crop improvement aspects in sorghum as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc& hybrid seed production	1
4	Crop improvement aspects in bajra as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc& hybrid seed production	1
5	Crop improvement aspects in urd, mung and cowpea as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc	1
6	Crop improvement aspects in pigeonpea as mentioned in the syllabus	1

	such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc& hybrid seed production	
7	Crop improvement aspects in soybean as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
8	Crop improvement aspects in sesame as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
9	Crop improvement aspects in groundnut as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
10	Crop improvement aspects in cotton and castor as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
11	Crop improvement aspects in chilli as mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
12	Crop improvement aspects in tomato mentioned in the syllabus such as Centers of origin, distribution of species Floral biology breeding objectives and procedures etc.	1
13	Modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional)	1
14	Seed production technology in self pollinated, cross pollinated and vegetatively propagated crops	1
15	Ideotype concept	1
16	Climate resilient crop varieties for future.	1

Lecture Schedule: Practical

S.N.	Topic	No. of lectures
1	Emasculation and hybridization techniques in rice, maize	1
2	Emasculation and hybridization techniques in sorghum and bajra	1
3	Emasculation and hybridization techniques in urd, mung, cowpea, pigeonpea	1

4	Emasculation and hybridization techniques in, soybean, sesame	1
5	Emasculation and hybridization techniques in and groundnut and cotton	1
6	Maintenance breeding of different kharif crops	1
7	Handling of germplasm and segregating populations by different methods like pedigree, bulk and single seed decent methods	1
8	Study of field techniques for seed production and hybrid seeds production in <i>Kharif</i> crops	1
9	Estimation of heterosis, inbreeding depression and heritability	1
10	Layout of field experiments	1
11	Study of quality characters	1
12	Donor parents for different characters	1
13	Visit to seed production plots	2
14	Visit to AICRP plots of different field crops	2

Suggested Readings

1. Breeding field crops -I by V.L. Chopra
2. Genetic improvement of field crops by C.B. Singh and D. Khare
3. Genetics and Breeding of Pulse crops by D.P. Singh
4. Vegetable breeding – Principles and Practices by Hari Har Ram
5. Breeding field crops by D.A. Sleper and J.M. Poehlman
6. Plant Breeding –theory and practice by S.K. Gupta
7. Breeding Asian field crops by J.M. Poehlman and D.N. Barthakur
8. Practical manuals on Crop Improvement I (*Kharif* crops) by Rajendra Kumar Yadav

HORT-311	Ornamental Crops, MAP and Landscaping	2(1+1)
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Objectives

1. To educate in detail about origin, area, climate, soil, improved varieties production technology of flowers and MAPs
2. To educate about concept, designing principles and components of landscaping
3. To educate about the physiological disorders of commercial flowers
4. To educate about the post-harvest management and value addition in flower crops and MAP.

Theory

Production technology of ashwagandha, costus, isabgol and geranium; Production technology of mint, aloe and ocimum, Coleus, Glory lily, Periwinkle etc.; Production technology of plants like lemongrass, citronella, vetiver and palmarosa etc., Importance and scope of ornamental crops;

Importance and scope of medicinal and aromatic plants and landscaping; Principles of landscaping; Landscape uses of trees, shrubs and climbers. Production technology of important cut flowers like rose, gerbera and orchids; Production technology of gladiolus, tuberose and lily; Production technology of chrysanthemum and carnation; Package of practices for loose flowers like marigold and jasmine under open conditions; Brief concept of home landscaping, Carpet bedding, Topiary, Bonsai, Lawn, flower arrangement, Herbaceous border, Hedge, Edge etc.; Processing and value addition important ornamental crops; Processing and value addition of MAPs produce.

Practical

Identification MAPs and Ornamental plants (trees, shrubs, climbers, seasonal flower and house plants). Propagation of MAP, Bed preparation and planting of MAP; Nursery bed preparation and sowing of seasonal flower seeds; Propagation of ornamental plants by terminal/herbaceous cuttings; Propagation of Anthurium and orchids; Propagation of bougainvillea; Planting of gerbera suckers; Gladiolus corms; Establishment and maintenance of lawn; Preparation of flower preservatives and their use in extending the vase life of cut flowers; Training and pruning of ornamental plants and raising of hedge and edge; Planning and layout of garden.

Lecture schedule: Theory

S. No.	Topics	No. of lectures
1.	Production technology of Ashwagandha, Costus, Isabgol and Geranium;	1
2.	Production technology of Mint, Aloe and Ocimum, Coleus, Glory lily, Periwinkle etc.;	1
3.	Production technology of plants like Lemongrass, Citronella, Vetiver and Palmarosa etc.,	1
4.	Importance and scope of ornamental crops and landscaping	1
5.	Importance and scope of medicinal and aromatic plants	1
6.	Principles of landscaping	1
7.	Landscape uses of trees, shrubs and climbers	1
8.	Production technology of important cut flowers like Rose, Gerbera and Orchids	2
9.	Production technology of Gladiolus, Tuberose and Lily	1
10.	Production technology of Chrysanthemum and Carnation	1
11.	Package of practices for loose flowers like Marigold and Jasmine under open conditions	1
12.	Brief concept of Home landscaping, Carpet bedding, Topiary, Bonsai,	2

	Lawn, flower arrangement, Herbaceous Border, Hedge, Edge etc.	
13.	Processing and value addition important ornamental crops	1
14.	Processing and value addition of MAPs produce	1
15.		16

Lecture schedule: Practical

S. No.	Topics	No. of lectures
1.	Identification MAPs and Ornamental plants (trees, shrubs, climbers, seasonal flower and house plants).	2
2.	Propagation of MAP	1
3.	Bed preparation and planting of MAP	2
4.	Nursery bed preparation and sowing of seasonal flower seeds	2
5.	Propagation of ornamental plants by terminal/herbaceous cuttings	1
6.	Propagation of Anthurium, orchids and bougainvillea	1
7.	Planting of gerbera suckers and Gladiolus corms	1
8.	Establishment and maintenance of lawn	1
9.	Preparation of flower preservatives and their use in extending the vase life of cut flowers	2
10.	Training and pruning of ornamental plants	1
11.	Raising of hedge and edge	1
12.	Planning and layout of garden	1
		16

Suggested readings

1. Floriculture in India by G.S. Randhawa and Mukopadhyay
2. Introduction to spices, plantation crops, medicinal and aromatic plants by N. Kumar, Abdul Khadder, P. Rangaswamy, I. Irulappam
3. Textbook of floriculture and landscaping by Anil K. Singh and Anjana Sisodia
4. Commercial flowers (Vol 1 and 2) by T.K. Bose
5. Yadav, P. K. & R. P. Singh 2021. Flower Production & Gardening, New India Publishing Agency, New Delhi, ISBN 9789390591213

PPATH-311	Diseases of Field & Horticultural Crops and their Management	3(2+1)
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Objectives

1. To study the symptoms produced on the host
2. To study the etiology of the diseases
3. To know about the disease cycle of the pathogens during pathogenesis
4. To study the epidemiological factors responsible for disease development
5. To study the management techniques for curbing the major diseases of field and horticultural Crops

Theory

Symptoms, etiology, disease cycle, epidemiology and management of major diseases of the following field and horticultural crops: Field crops- Rice (blast, brown spot, sheath blight, false smut, bacterial leaf blight, bacterial leaf streak, tungro, khaira); Wheat (rusts, loose smut, Karnal bunt); Maize (banded leaf and sheath blight, southern and northern blight, downy mildew); Sorghum (smuts, grain mold, anthracnose); Bajra (downy mildew, ergot) and Finger millet (blast, leaf spot); Groundnut (early and late leaf spots, rust, wilt); Soybean (rhizoctonia blight, bacterial spot, seed and seedling rot, mosaic); Grams (*Ascochyta* blight, wilt, grey mold); Pea (downy mildew, powdery mildew, rust); Black gram and Green gram (web blight, *Cercospora* leaf spot, anthracnose, yellow mosaic); Sugarcane (red rot, smut, grassy shoot, ratoon stunting, PokahBoeng); Mustard (*Alternaria* blight, white rust, downy mildew, sclerotinia stem rot) and Sunflower (sclerotinia stem rot, *Alternaria* blight); Cotton (anthracnose, vascular wilts, blackarm).

Horticultural crops: Citrus (canker, gummosis) and Guava (wilt, anthracnose); Banana (sigatoka, Panama wilt, bacterial wilt, bunchy top); Papaya (foot rot, leaf curl, mosaic) and Pomegranate (bacterial blight); Apple (scab, powdery mildew, fire blight, crown gall) and Peach (leaf curl); Grapevine (downy mildew, powdery mildew, anthracnose) and Strawberry (leaf spot); Coconut (bud rot, *Ganoderma* wilt), Tea (blister blight) and Coffee (rust); Mango (anthracnose, malformation, bacterial blight, powdery mildew); Potato (early and late blight, black scurf, leaf roll, mosaic) and Tomato (damping off, wilt, early and late blight, leaf curl, mosaic); Brinjal (phomopsis blight and fruit rot, sclerotinia blight) and Chilli (anthracnose and fruit rot, wilt, leaf curl); Cucurbits (powdery and downy mildew, wilts) and Cruciferous vegetables (*Alternaria* leaf spot, black rot, cauliflower mosaic); Beans (anthracnose, bacterial blight) and Okra (yellow vein mosaic); Ginger (soft rot), Turmeric (leaf Spot) and Coriander (stem gall); Rose (dieback, powdery mildew, black leaf spot) and Marigold (*botrytis* blight, leaf spots).

Practical

To study the symptoms of different diseases of field and horticultural crops: Blast and brown spot of rice, sheath blight and bacterial leaf blight of rice, downy mildew and powdery of cucurbits, rhizoctonia and *Cercospora* leaf spot of green gram / black gram, *Alternaria* blight and downy mildew of mustard, early blight of late blight of potato and tomato, Phomopsis blight of brinjal, powdery mildew and rust of pea, stem gall of coriander, anthracnose and fruit rot of chilli, taphrina leaf spot of turmeric, red rot of sugarcane, acquaintance with fungicides, antibiotics and biopesticides and their use in management of diseases of horticultural crops. Identification and histopathological studies of selected diseases of field and horticultural crops covered in theory.

Field visit for the diagnosis of field problems, Collection and preservation of plant diseased specimens for herbarium.

Lecture Schedule: Theory

S.No	Topic	No. of lectures
1.	Rice : Blast, brown spot, sheath blight, false smut, bacterial leaf blight, bacterial leaf streak, tungro and khaira	02
2.	Wheat:Rusts, loose smut, Karnal bunt and ear cockle &tundu	02
3.	Maize: Banded leaf and sheath blight, southern and northern blight and downy mildew	01
4.	Sorghum: Smuts, grain mold and anthracnose.	01
5.	Bajra: Downy mildew and ergot. Finger millet: Blast and leaf spot	01
6.	Groundnut: Early and late leaf spots, rust and wilt	01
7.	Soybean: Rhizoctonia blight, bacterial spot, seed and seedling rot and mosaic	01
8.	Grams: Ascochyta blight, wilt and grey mold	01
9.	Pea: Downy mildew, powdery mildew and rust	01
10.	Black gram and Green gram: Web blight, Cercospora leaf spot, anthracnose and yellow mosaic	01
11.	Sugarcane: Red rot, smut, grassy shoot, ratoon stunting and Pokah Boeng	01
12.	Mustard: Alternaria blight, white rust, downy mildew, sclerotinia stem rot. Sunflower Sclerotinia stem rot and Alternaria blight.	02
13.	Cotton: Anthracnose, vascular wilts and blackarm	01
14.	Citrus: Canker and gummosis. Guava: Wilt and anthracnose	01
15.	Banana: Sigatoka, Panama wilt, bacterial wilt and bunchy top	01
16.	Papaya: Foot rot, leaf curl and mosaic. Pomegranate: bacterial blight	01
17.	Apple: Scab, powdery mildew, fire blight and crown gall and Peach: leaf curl	01
18.	Grapevine: Downy mildew, powdery mildew, anthracnose. Strawberry: leaf spot	01
19.	Coconut: Bud rot and Ganoderma wilt. Tea: Blister blight Coffee: Rust	01
20.	Mango: Anthracnose, malformation, bacterial blight and powdery mildew	01
21.	Potato: Early and late blight, black scurf, leaf roll and mosaic	02
22.	Tomato: Damping off, wilt, early and late blight, leaf curl and mosaic	01
23.	Brinjal: Phomopsis blight and fruit rot and sclerotinia blight	01
24.	Chilli : Anthracnose and fruit rot, wilt and leaf curl	01

25.	Cucurbits: Powdery and downy mildew and wilts. Cruciferous Vegetables: Alternaria leaf spot, black rot and cauliflower mosaic	01
26.	Beans: Anthracnose and bacterial blight. Okra: Yellow vein mosaic	01
27.	Ginger: Soft rot. Turmeric: leaf Spot. Coriander: Stem gall	01
28.	Rose: Dieback, powdery mildew and black leaf spot. Marigold: Botrytis blight and leaf spots).	01

Lecture Schedule: Practical

S.No	Topic	No. of lectures
1.	Wheat: Rusts, loose smut, karnal bunt and ear cockle.	01
2.	Gram: Root rot, wilt and Ascochyta blight.	01
3.	Bajra: Downy mildew and ergot.	01
4.	Field visit to diagnose the diseases and collect of disease specimen	01
5.	Maize: leaf blights and downy mildews.	01
6.	Sorghum: Grain smut and anthracnose.	01
7.	Moong, urd and moth beans: Web blight and yellow mosaic.	01
8.	Groundnut: Tikka, collar rot and peanut clump virus.	01
9.	Okra: Yellow vein mosaic. Date palm: Graphiola leaf spot	01
10.	Isabgol: Downy mildew. Cumin: Wilt, powdery mildew and Alternaria blight.	01
11.	Potato: Late blight, black heart Onion: Purple blotch. Chillies: Anthracnose and leaf curl.	01
12.	Cucurbits: Powdery mildew, mosaic, Choanephora rot and root knot	01
13.	Field visit to diagnose the diseases and collect of disease specimen	01
14.	Cabbage: Alternaria leaf spot and black rot. Pea: powdery mildew.	01
15.	Brinjal: Little leaf, Tomato: Damping off, early blight, leaf curl and root knot.	01
16.	Acquaintance with fungicides, antibiotics and biopesticides and their use in management of diseases of horticultural crops	01

Suggested Readings

1. Integrated Plant Disease Management by R.C. Sharma
2. Plant Diseases by R.S. Singh
3. Plant Disease Management: Principles and Practices by Hriday Chaube
4. Plant Pathology by G.N. Agrios
5. Diseases of Vegetables Crops by Y. K. Gupta and Y. S. Paul

PPHYS-311	Fundamentals of Crop Physiology	3(2+1)
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Theory:

Definitions of plant physiology and crop physiology, Importance of crop physiology, Relationship of crop physiology with other branches of crop science

Diffusion and osmosis, Physiological roles of water to crop plants, Definition of water potential and components of water potential, Water absorption by plants: Concept of active and passive absorption, Water loss by plants: Types of water loss: transpiration, stomatal physiology and guttation, Water use efficiency and factors affecting WUE.

Classification of mineral elements: Essential and beneficial elements, Passive and active transport of mineral element: Nernst equation, ion transport, ion pump and channels, Functions of essential elements and their deficiency and toxicity symptoms, Hydroponics and sand culture.

Brief outline of: Photosynthetic apparatus, pigment system, quantum requirement and quantum yield: Structure of chloroplast, Examples of different photosynthetic pigments (chlorophyll, carotenoids, phycobilins etc.), Difference between chlorophyll a and chlorophyll b, Structure of chlorophyll a and chlorophyll b, Short discussion on quantum requirement and quantum yield, Red drop and Emerson enhancement effect, Pigment system I & II.

Introduction to light reaction of photosynthesis, Light absorption by photosynthetic pigments and transfer of energy, Source of O₂ during photosynthesis: Hill reaction, Brief introduction to cyclic and non-cyclic photo-phosphorylation: production of assimilatory powers.

Introduction to C₃ C₄ and CAM pathways: Calvin Cycle, Hatch & Slack Cycle, CAM Cycle, Significance of these pathways (concept of photorespiration, absence of photorespiration in C₄ plant: Productivity of C₄ plant, CAM: an adaptive mechanism), Factors affecting photosynthesis (light, temperature, CO₂ O₂ etc). Outline of the process of respiration: Definition and importance, Glycolysis, Krebs Cycle and ETC, Factors affecting respiration O₂ temperature, CO₂ etc.).

Terminologies/Definitions: Growth, Development and Differentiation. Measurement of plant growth (fresh weight, dry weight, linear dimension, area etc). Introduction to CGR, RGR, NAR etc. Short discussion on factors affecting growth and development.

Photoperiodism: Classical works of Garner and Allard. Photoperiodic Classification of plants: Short Day Plant, Long Day Plant, Day Neutral plant etc. Introduction to Photoperiodic induction Site of photo-inductive perception, Role of Phytochrome Introduction to Vernalization (What is vernalization, devernialization etc). Meaning, classification (seasonal, sequential etc.), relation with abscission. Physiological and biochemical changes during senescence, Abscission and its significance, Concept of stay green, Hormonal regulation of senescence.

Terminologies / Definitions: Plant hormone, Plant growth regulators (PGR), Plant growth inhibitor. Recognized classes of PGR (Auxins, Gibberellins, Cytokinins, Ethylene and Absciscic

acid) and their major physiological roles, Agricultural uses of PGRs (IBA, NAA, 2, 4-D, GAs, Kinetin etc).

Practical:

Study on structure and distribution of stomata; Demonstration of imbibition, osmosis, plasmolysis, estimation of water potential, relative water content; Tissue test for mineral nutrients, identification of nutrient deficiency and toxicity symptoms in plant; Estimation of photosynthetic pigments, rate of photosynthesis, respiration and transpiration; Plant growth analysis; Study on senescence and abscission, hormonal regulation of senescence; Demonstration of the effects of different PGRs on plants.

Theory Schedule

Class	Topic
1	Introduction: Definitions of plant physiology and crop physiology; Importance and scope.
2	Relationship of crop physiology with other crop sciences.
3	Diffusion and osmosis; Physiological roles of water in crop plants.
4	Definition and components of water potential; Active and passive absorption of water.
5	Water loss in plants: Transpiration, stomatal physiology, and guttation.
6	Water use efficiency (WUE) and factors affecting WUE.
7	Classification of mineral elements: Essential and beneficial elements.
8	Transport of mineral elements: Nernst equation, ion transport, ion pump, and channels.
9	Functions of essential elements; Deficiency and toxicity symptoms; Hydroponics and sand culture.
10	Photosynthetic apparatus: Structure and function of pigments; Chlorophyll a and b.
11	Light reactions: Quantum requirement, quantum yield, and pigment systems I & II.
12	Cyclic and non-cyclic photophosphorylation; Hill reaction; Oxygen evolution in photosynthesis.
13	C ₃ , C ₄ , and CAM pathways: Calvin Cycle, Hatch & Slack Cycle, and CAM Cycle.
14	Photorespiration and its significance; Productivity of C ₄ plants and CAM adaptations.
15	Factors affecting photosynthesis: Light, temperature, and CO ₂ levels.
16	Respiration: Overview of glycolysis, Krebs Cycle, and Electron Transport Chain (ETC).
17	Factors affecting respiration: Oxygen, temperature, and CO ₂ .
18	Growth, development, and differentiation: Definitions and measurements.
19	Growth parameters: CGR, RGR, NAR, and factors affecting growth and development.
20	Photoperiodism: Classical works of Garner and Allard; Photoperiodic classification.
21	Photoperiodic induction, phytochrome, and site of photo-inductive perception.
22	Vernalization: Definition, classification, and significance.
23	Senescence: Classification, physiological, and biochemical changes.
24	Abscission: Process, significance, and concept of "stay green."
25	Plant hormones and growth regulators (PGR): Definitions and classifications.
26	Physiological roles of Auxins and Gibberellins.
27	Physiological roles of Cytokinins, Ethylene, and Absciscic acid.

Class	Topic
28	Agricultural applications of PGRs (e.g., IBA, NAA, 2,4-D, GA ₃ , Kinetin).
29	Recap of water relations, mineral nutrition, and photosynthesis.
30	Recap of respiration, growth, and development.
31	Recap of photoperiodism, vernalization, senescence, and abscission.
32	Recap of plant hormones, growth regulators, and their applications.

Practical Schedule

Class	Topic
1	Study the structure and distribution of stomata.
2	Demonstration of imbibition and osmosis.
3	Demonstration of plasmolysis and estimation of water potential.
4	Estimation of relative water content in plant tissues.
5	Tissue test for identifying mineral nutrient deficiencies.
6	Identification of nutrient deficiency and toxicity symptoms in plants.
7	Estimation of photosynthetic pigments (chlorophyll a, b, and carotenoids).
8	Measurement of the rate of photosynthesis under varying conditions.
9	Estimation of respiration rate in plant tissues.
10	Measurement of transpiration using a potometer.
11	Plant growth analysis: Fresh weight, dry weight, and linear dimensions.
12	Study of senescence and abscission in plants.
13	Demonstration of hormonal regulation of senescence.
14	Effects of plant growth regulators (e.g., IBA, NAA, GA ₃) on plant growth.
15	Effects of plant growth inhibitors (e.g., ABA) on physiological processes.
16	Final review: Integration of practical observations with theoretical concepts.

Suggested readings:

1. Plant Physiology by Robert M. Devlin and Francis H. Witham
2. Plant Physiology by Lincoln Taiz and Eduardo Zeiger
3. Plant physiology by Frank B. Salisbury and Cleon W. Ross
4. Fundamentals of Plant Physiology by Lincoln Taiz, Eduardo Zeiger, Ian Max Melle and Angus Murphy
5. Devlin's Exercises in Plant Physiology by Robert Devlin, Francis H. Witham and David F. Blaydes

ET-311	Educational Tour**	2(0+2)
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